

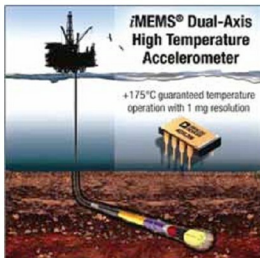
substrate leakage current. Some others are decreased carrier mobility, variation in device parameters such as threshold voltage, beta and saturation voltage, increased electro-migration of metal interconnects, and decreased dielectric breakdown strength," says Choudhury.

Let us look at how some of these problems are overcome in new technologies.

Handling substrate leakage. Although standard silicon can operate well beyond the military requirement of 125°C, leakage in standard silicon processes doubles for every 10°C increase in temperature, making it unacceptable for many precision applications. Trench isolation, silicon-on-insulator (SOI) and other variations to the standard silicon process help decrease leakage and enable high-performance operation at temperatures above 200°C. Use of wide-band-gap materials, such as silicon carbide (SiC), helps raise the limits even further, to as high as 600°C.

Testing and characterisation at high temperatures. Simulating extreme temperature environments in the lab to verify, qualify and characterise the components can be very challenging. Special materials, ovens and chambers are required to create the test platform, wherein the tests have to be done carefully without damaging the printed circuit board (PCB), by gradually increasing or decreasing the temperature. Specialised tools are available today, which can operate at such high temperatures to test and control electronic systems.

"Special industrial, aerospace and defence applications like monitoring oil and gas pipelines, and controlling heavy machinery and space vehicle test systems require control units and sensors that can function even at extreme conditions of temperature, pressure and vibration. These critical applications also demand extremely low failure rates. Testing the sensors and control systems thoroughly for all environmental conditions is necessary before they can be deployed in the field. Special test set-up with rugged



ADI ADXL206 accelerometer

control and measurement systems that can endure similar conditions are required to validate the functioning of these sensors and control units," says Ravichandran Raghavan, technical marketing engineer, National Instruments, India (NI). NI has solutions for design and test of extreme temperature components.

Packaging the die. While plastic packaging would withstand temperatures up to 175°C, other technologies would be needed beyond that. TI, for example, offers three types of packages to support high-temperature applications: plastic package for 175°C, ceramic and known good die (KGD) packaging for higher temperatures. Ceramic and KGD parts are characterised to 210°C, operate in an extended temperature range of -55°C to +210°C, and have a high-temperature operating life of 1000 hours. Even the plastic package is ruggedised using special techniques to improve reliability at high temperatures.

"While packaging, care must be taken to match the coefficient of thermal expansion between die, die-attach and substrate, so that the die is not stressed or fractured over cycles of wide temperature span," suggests Choudhury.

Managing circuit design considerations: "Those designing circuits for high-temperature use must account for changes in IC parameters and passive components over a wide temperature range, paying close attention to their behaviour at the temperature extremes

to ensure circuit operation within the target limits," says Choudhury. They must analyse offset and input bias drift, gain errors, temperature coefficients, voltage ratings, power dissipation, board leakage and intrinsic leakage of other discrete devices. Often, high-temperature operation worsens board leakages due to contaminants such as solder flux, dust or condensation. Designers usually overcome such problems by using a proper layout that ensures adequate spacing between sensitive nodes.

Being put to good use

Using special techniques and engineering skills as discussed above, chip makers now offer full extreme temperature silicon chains for various verticals. Several components are available today, which are guaranteed to withstand extreme temperatures. Examples include high-precision temperature sensors (ranging from 1 to 0.25-degree accuracy), low-voltage programmable thermostatic switches, airflow and liquid flow sensors, digitally-programmable sensor signal amplifiers, instrumentation amplifiers, high-G micro-electromechanical accelerometers and gyroscopes, micro-processor supervisory circuits, signal processors, power management and data conversion systems, test instrumentation, and more. Let us take a look at some of these products available in the market.

Accelerometers. High-temperature accelerometers are available for vibration, speed and gravity measurement in industrial applications, geological down-hole exploration, etc. One such example is ADI's ADXL206—a high-precision, low-power, dual-axis accelerometer that uses its integrated micro-electromechanical system (iMEMS) technology to integrate the sensor and signal-conditioned voltage outputs on a single, monolithic IC. It can measure both dynamic (vibration) and static acceleration (gravity) with an accuracy of 1mg resolution at 60 Hz. It is housed in an eight-lead side-brazed